
Open and closed innovation – different innovation cultures for different strategies

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Abstract: Although strategies, processes, or the role of business models have been addressed in the open innovation literature, the people side of the equation – i.e., the underlying innovation culture – has been neglected so far. Whereas cultural requirements of open innovation have been mentioned, such as the need to overcome the not-invented-here (NIH) syndrome, there exists, to the best of our knowledge, no study that empirically examines open innovation cultures. We attempt to fill this research gap by focusing on innovation cultures within three business units (two follow a closed and one follows an open innovation approach) of a leading multinational company within the specialty chemicals industry. Employing an overall sample of 109 respondents, we focus on the cultural dimensions of NIH syndrome, risk-taking, and management support of innovative behaviour and provide first evidence of cultural differences between Open and Closed Innovation units.

Keywords: open innovation; closed innovation; innovation culture; corporate culture; innovation strategy; not-invented-here syndrome; NIH; management support; risk-taking; chemical industry.

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1 Introduction

Firms acting in an environment of rapid technological change are often depending on externally developing knowledge sources in order to generate radical innovations. This pressing need of integrating external R&D sources has forced many firms to shift from a closed innovation model to an open innovation model (Chesbrough, 2003). Open innovation denotes on the one hand, the use of external and internal knowledge sources to accelerate internal innovation, and on the other hand, the use of external paths to markets for internal knowledge. However, many of the open innovation tools, such as licensing (e.g., Tidd and Trewhalla, 1997; Arora et al., 2002; Atuahene-Gima and Patterson, 1993), joint R&D agreements (e.g., Ritter and Gemünden, 2003; Arora and Gambardella, 1990; Steensma and Corley, 2000), minority investments and corporate venture capital (e.g., Dushnitsky and Lenox, 2005; Ernst et al., 2005; Markham et al., 2005), or spin-offs (e.g., Lichtenthaler, 2006) were well-known before the term took root in theory and practice. But open innovation is more than just the sum of its parts. Open innovation is a holistic approach to innovation management as “systematically encouraging and exploring a wide range of internal and external sources for innovation opportunities, consciously integrating that exploration with firm capabilities and resources, and broadly exploiting those opportunities through multiple channels” [West and Gallagher, (2006), p.320].

Since open innovation is a complex issue, various research streams, such as outsourcing of R&D, globalisation of innovation, early supplier and user integration, external technology commercialisation and application, or the role of the business model, have contributed to the development of the field (Gassmann, 2006). Furthermore, open innovation can and should be analysed at a number of different levels, such as individuals, firms, dyads, interorganisational networks, and nations or regions. Such an approach which distinguishes between different levels of analysis can broaden the scope and deepen the understanding of open innovation (Vanhaverbeke, 2006; Vanhaverbeke and Cloudt, 2006). Although various research streams have contributed to research on open innovation and some of the different levels of analysis have been addressed in previous studies, the evolving debate is missing a key element: the people side of the equation. This seems paradoxical since many firms are facing difficulties during the implementation of the concept. In a recent survey by Bain & Company (2007), open innovation received a satisfaction rating of only 3.7 out of 5, ranking it 15th out of 25 management tools. Furthermore, the trend in corporate practice towards opening up the firm's boundaries to outside innovation seems to continue. As the latest annual survey of the Industrial Research Institute (IRI) on R&D trends forecast revealed, firms expect a significant increase in the use of outside resources (Scinta, 2008). To put it another way: open innovation will continue being of major importance in corporate practice for innovation management. Therefore, firms need to overcome problems in coping with open innovation. According to Docherty (2006), many firms pay too much attention on external ideas and technologies or the processes to identify them. Although those technologies and processes are necessary, “[i]t's just as important to focus on the interpersonal, cultural, and implementation challenges” [Docherty, (2006), p.15].

With regard to the different levels of analysis, much of the open innovation research has focused on the firm level. The first level of analysis – the level of individuals – has

not received much attention in previous research on open innovation. Since many firms seem to struggle leveraging the commercial potential of technologies that have been developed outside the firm boundaries, Vanhaverbeke (2006, p.207) states that “it is interesting to analyse how firms’ internal organisation plays a role in improving the assessment and integration of externally acquired knowledge.” The same applies to the external commercialisation of internal knowledge and technologies. Furthermore, as the abovementioned definition implies, open innovation requires a different way of thinking and a change in employees’ practices in dealing with ideas, knowledge, or technologies. With regard to the research streams contributing to the field of open innovation, a change in a firm’s innovation culture therefore needs to accompany the move towards open innovation. However, little is known about innovation cultures in open innovation settings. Whereas cultural requirements of open innovation have been mentioned in the literature, such as the need to overcome the not-invented-here (NIH) syndrome (Chesbrough, 2003), there exists – to the best of our knowledge – no study that empirically examines open innovation cultures. Therefore, the purpose of this paper is to fill this white spot in research on open innovation by addressing the following research questions:

- 1 What are the special requirements for open innovation cultures compared to closed innovation cultures, i.e., do both innovation cultures need to be different?
- 2 If innovation cultures within open and closed innovation environments need to be different, which are some key cultural dimensions these differences refer to and is this supported by empirical evidence?

The paper proceeds as follows. In the following section, innovation culture is defined and its constituent dimensions are presented. Based on this discussion, the theory of open innovation, and the resource-based view (RBV) of the firm, hypotheses on the cultural dimensions NIH syndrome, risk-taking, and management support are outlined. The following empirical sections describe the chemical industry, which is the research setting of this study, and the data sources used to test our hypotheses. Then, results are presented and discussed. Finally, managerial implications are outlined in a concluding section.

2 Innovation culture

Although literature lacks consensus about the definition of corporate culture, there is broader agreement that a major aspect of culture is that groups share or hold certain things in common (Schein, 1997). These things are usually referred to as values, norms, attitudes, and behaviour patterns, etc., that form the core identity of an organisation or of its sub-units. These “shared values and beliefs .. help individuals understand organisational functioning and thus provide them norms for behavior in the organisation” [Deshpandé and Webster, (1989), p.4]. Accordingly, three different and highly interrelated levels of culture can be distinguished and analysed. They include

- 1 shared basic assumptions or values
- 2 behavioural norms
- 3 artefacts and behaviours (Schein, 1997; Homburg and Pflesser, 2000).

These levels refer to different degrees of a culture's visibility to the observer. Artefacts include rituals and ceremonies, stories, arrangements, and language created by an organisation. Together with corporate behaviour patterns, they build the surface level, i.e., the most visible level of organisational culture (Schein, 1997; Homburg and Pflesser, 2000). Hofstede et al. (1990) subsume these visible elements under the term 'practices'. Behavioural norms can be viewed as social principles, goals, philosophies, and standards that define appropriate attitudes and legitimate specific behaviours (O'Reilly and Chatman, 1996). They are further much more specific and relevant for these behaviours than values (Katz and Kahn, 1978; Hatch, 1993; Homburg and Pflesser, 2000). As the deepest manifestation of culture, values can be characterised as broad and non-specific feelings of what is, for instance, positive and negative, good and bad, normal and abnormal, or important and unimportant (Hofstede et al., 1990; de Brentani and Kleinschmidt, 2004).

An important subculture is the firm's or business unit's (BU) innovation culture. Taking into account the different cultural levels, we define innovation culture as

- 1 organisation-wide shared basic values that support innovation
- 2 organisation-wide norms for innovation
- 3 perceptible innovation-oriented practices.

The magnitude of research focuses on specific aspects of innovation culture. For example, Hurley and Hult (1998) focus on power sharing, participative decision-making, as well as on learning and development. Others like de Brentani and Kleinschmidt (2004) concentrate on management support in terms of encouraging entrepreneurship, new ideas, and failure tolerance. Overall, most empirical studies find a strong innovation culture being closely related with innovative performance (e.g., Capon et al., 1992; Cooper and Kleinschmidt, 1995; Hurley and Hult, 1998; de Brentani, 2001; Ernst, 2003; de Brentani and Kleinschmidt, 2004).

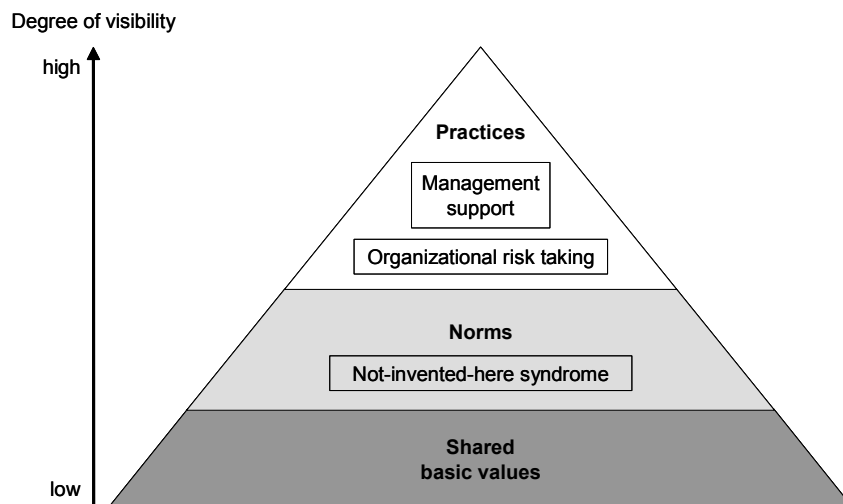
However, the question arises of what are the values, norms, and practices that are characteristic for such a strong innovation culture. For example, are there certain values or norms that are more likely to support innovation than others? Thus, innovation – although wanted by the firm – may not be possible if values, norms, and practices emphasise the status-quo. In addition to the aforementioned empirical studies, many elements of an innovation culture can be found in related orientation constructs, such as market orientation (Deshpandé et al., 1993; Slater and Narver, 1995; Hult et al., 2004), technology orientation (Gatignon and Xuereb, 1997; Zhou et al., 2005), entrepreneurial orientation (Lumpkin and Dess, 1996; Hult et al., 2004), or learning orientation (Hult et al., 2004; Kandemir and Hult, 2005), that have been linked to innovation and regarded as part of an organisation's culture. Although literature lacks a common definition of innovation culture, different conceptualisations are characterised by some communalities of the underlying components of such a culture. Usually, literature refers to innovation culture as a firm's or BU's internal environment that, for instance, encourages risk-taking (e.g., Ernst, 2003; de Brentani and Kleinschmidt, 2004; Kleinschmidt et al., 2007), supports openness to new ideas (e.g., Zaltman et al., 1973; van de Ven, 1986; Hurley and Hult, 1998), tolerates failures (e.g., van de Ven and Chu, 1989; Sundgren et al., 2005), fosters learning (e.g., Hurley and Hult, 1998; Sundgren et al., 2005), and promotes constructive dissent (e.g., Capon et al., 1992; van de Ven and Chu,

1989; Day and Schoemaker, 2000). All previous empirical studies on innovation culture implicitly follow the corporate culture paradigm, which views culture as an internal variable of the organisation that can be shaped by management to pursue strategic goals (Smircich, 1983).

With regard to the RBV of the firm (Penrose, 1959; Wernerfelt, 1984), innovation culture is a valuable resource of a firm. It evolves over time, is intangible and difficult to imitate, and might lead to competitive advantage (Kleinschmidt et al., 2007; Barney, 1986). According to Barney (1986), a culture is only valuable when it fits the strategic and competitive context. Hence, an innovation culture must fit the innovation strategy followed. Therefore, the innovation strategies open innovation and closed innovation may be different regarding their cultural requirements. First evidence can be found in the literature arguing that employees of an open innovation organisation need to be much more adaptive than their counterparts applying a closed innovation approach (Chesbrough, 2003). This becomes even more important in case of more complex and more turbulent environments. As Chesbrough (2003) noted, many firms shift their innovation model from a closed innovation approach to an open innovation approach. However, what would such a shift imply for the underlying innovation culture? Although a firm may want to shift its innovation strategy from closed to open, it may face difficulties to overcome its cultural path dependencies. The requirement that a valuable innovation culture must fit the strategic context implies that a firm's culture has fitted the closed innovation strategy, but may not be appropriate anymore for the open innovation strategy. Hence, a firm needs to be able to successfully modify its culture, i.e., it must have the necessary culture management skills (Barney, 1986). According to Teece et al. (1997), culture cannot be acquired; it rather has to be built. This refers to the notion of dynamic capabilities enabling a firm to adapt its intangible resource 'innovation culture' to its innovation strategy. Thus, to successfully shift from closed to open innovation, firms need the dynamic capabilities to modify their innovation culture.

3 Hypotheses

Although firms need to modify their innovation cultures, the question remains, which cultural aspects require special attention during the transformation process? In the following, three selected dimensions of an innovation culture are discussed. For each cultural dimension, it is argued if this element needs to be different in an open innovation environment compared with a closed innovation environment. The hypotheses not only address selected elements of an innovation culture as identified in the previous chapter but also refer to the upper two levels of innovation culture, i.e., norms and practices. Openness to new ideas is addressed by taking into account employees' attitudes towards external technology sourcing. Thus, only openness to external ideas and technologies is considered here. The underlying attitude that will be discussed is the NIH syndrome that generally denotes a negative attitude towards external technology sourcing. This attitude is considered belonging to the level of norms. Furthermore, the surface level of innovation culture, i.e., the level of observable practices, will be addressed. Here, we focus on organisational risk-taking and general management support of innovative behaviour. The different aspects of innovation culture that will be discussed in the following and their respective cultural levels are illustrated in Figure 1.

Figure 1 Dimensions and levels of innovation culture analysed in this study

3.1 NIH syndrome

Since open innovation puts strong emphasis on the use of external technology within the innovation process, individual employees as well as the firm as a whole need to be sufficiently open towards external technology. Accordingly, it is critical for the success of an open innovation strategy to overcome certain attitudes that inhibit the firm to fully exploit its potential (Lichtenthaler and Ernst, 2006). Many scholars and practitioners have referred to the NIH syndrome to describe the negative effects resulting from an overemphasis on internal technologies, ideas, or knowledge (e.g., Katz and Allen, 1982; Cohen and Levinthal, 1990; Brockhoff, 1997; Leonard-Barton, 1998; Chesbrough, 2006; West and Gallagher, 2006). However, only a few works explicitly address and empirically analyse the NIH syndrome (Katz and Allen, 1982; Menon and Pfeffer, 2003). The probably most detailed and comprehensive work is provided by Mehrwald (1999). According to Mehrwald [1999, p.5, translation taken from Lichtenthaler and Ernst (2006)], the “NIH syndrome represents a negatively biased, invalid, generalising and rigid attitude of individuals or groups to externally developed technology, which may lead to an economically detrimental neglect or suboptimal use of external technology.” Studies on the NIH syndrome generally show consistent results. Regarding the consequences of being infected with the NIH syndrome, two major findings can be distinguished:

- 1 the biased, wrong and generalised evaluation of external technology
- 2 the negative impact on a firm’s innovation performance, which may finally result in the failure of the implementation of external technologies (Clagett, 1967; Katz and Allen, 1982; Mehrwald, 1999; Lichtenthaler and Ernst, 2006).

Since on average, Mehrwald (1999) found a neutral and even rather positive attitude to external technology among R&D managers and scientists, one could assume that the NIH syndrome is a theoretical concept that has no meaning in managerial practice. This

finding is supported by Menon and Pfeffer (2003) who provide evidence that firms may even prefer outsider knowledge over knowledge from internal sources. However, Mehrwald (1999) also identified some individuals (35.7%) who exhibited some warnings of being infected with the NIH syndrome. Being slightly infected with the NIH syndrome might not be problematic for an organisation following a closed innovation approach. However, because sourcing ideas, technologies and knowledge from the external environment is a major building block of a firm's open innovation strategy, being infected with the NIH syndrome would be disastrous for such a firm. Accordingly, Witzeman et al. (2006, p.27) propose that "[h]arnessing external technology for innovation requires a fundamental change in employee thinking. The 'Not Invented Here' syndrome is replaced with the 'Invented Anywhere' approach." Similarly, Huston and Sakkab (2007, p.23) state that the innovation culture has to move "from, 'not-invented-here' to one based on 'proudly-found-elsewhere.'" Therefore, we hypothesise:

Hypothesis 1 Open Innovation units and Closed Innovation units are culturally different in that Open Innovation units are less infected with the NIH syndrome than Closed Innovation units.

3.2 *Organisational risk-taking*

Furthermore, innovation, i.e., developing new products, technologies or processes and introducing them to the market, often comes with the risk of failure. Creating an innovation usually involves combining different resources. For instance, an innovation may require a new technology which is not currently available to the firm. The firm, then, has to decide if the technology is to be developed internally or to be sourced from outside the firm. Either way, this is usually associated with risk and experimentation (Atuahene-Gima and Ko, 2001). Following an open innovation strategy, however, involves higher risks in comparison to pursuing a closed innovation strategy. The reasons for these different risk levels will be discussed in the following with respect to external technology sourcing, since this is a major building block of an organisation's open innovation strategy.

Literature on innovation and technology management often cites the phenomenon of shortening product life cycles and resulting pressure to accelerate development times. Being risk averse and thus putting too many risk filters in place may dramatically slow down the flow of the innovation project (Smith and Reinertsen, 1992). This could result in missing the window of opportunity (Calantone et al., 2003). Nevertheless, participants in the innovation project seek to minimise project risks. Compared to team members' perception that internal technology sourcing poses enough risk to the innovation project, external technology may be perceived as even increasing project risk. As argued by Chesbrough (2006), this is due to higher uncertainties (when compared with internal technology sourcing) about the external source of a respective technology. Thus, although the expected value of an external technology may be even higher than the value of an internal technology, the variance around that expected value may be higher as well (Chesbrough, 2006). The project risk also increases when external technology sourcing is conducted in a joint effort with other organisations. In joint R&D agreements or joint ventures, for instance, innovation success depends on the joint probability that the participating partners will be able to fulfil their commitments (Adner, 2006).

When compared with internal technology sourcing, external technology sourcing may also come along with higher risk because of difficulties in assessing external technologies. These difficulties are of particular relevance in case of increasing technological complexity (Doering and Parayre, 2000; Tidd et al., 2005). One major underlying problem of assessing the performance or feasibility of an external technology refers to information asymmetries. According to Hauschildt (1992), the technology provider has a qualitatively higher level of knowledge about the technology than the technology recipient. This provides room for opportunistic behaviour, i.e., the technology provider may take advantage of the information asymmetry and may 'cheat' on the technology sourcing organisation (Ring and van de Ven, 1989). However, Conner and Prahalad (1996) argue that even when the technology provider has no intentions to be opportunistic, transaction costs may still be high. In this case, the risk of not being able to sufficiently assess technological feasibility can be attributed to the tacit nature of the technological knowledge to be transferred. Overall, a mistaken technology assessment may result in problems with regard to the fit with internal capabilities or the technology's manufacturability (Doering and Parayre, 2000). Therefore, we hypothesise:

Hypothesis 2 Open Innovation units and Closed Innovation units are culturally different in that Open Innovation units are more prone to taking risks than Closed Innovation units.

3.3 *Management support*

Studies of correlates of new product development (NPD) performance indicate a positive relationship between an innovation environment where management supports innovative behavior and NPD performance (Henard and Szymanski, 2001). One possibility to support innovation is to provide sufficient resources in terms of people, budget, and time. Thus, management needs to provide funding for innovation concerns, such as technology sourcing or hiring the right people for the innovation effort (de Brentani and Kleinschmidt, 2004). Besides those 'harder' forms of management support, management can also be personally involved in the innovation process. The value of management playing a visible, involved role during the innovation process has received great attention in research on innovation and technology management as well as in managerial practice (Chakrabarti and Hauschildt, 1989; Howell and Higgins, 1990; Hauschildt and Kirchmann, 2001). A variety of expressions has been used to refer to these informal, supporting roles. Among them, two of the most often applied ones are 'champion' (Chakrabarti, 1974; Howell and Higgins, 1990; Markham, 2000; Howell et al., 2005) and 'promotor' (Hauschildt and Gemünden, 1999; Hauschildt and Kirchmann, 2001). At its roots, the champion or promotor concept is based on the assumption that innovation usually comes up against resistance, which results in different types of conflict. To overcome these barriers to innovation, promotors or champions are required.

Although champions and promotors are of particular importance in specific innovation projects and to overcome project-specific barriers, management can also actively create such an innovation supportive environment. Buckler (1997) points out that top management needs to provide a climate in which everybody understands that producing new ideas and innovation is a vital necessity for the firm or BU. According to Amabile et al. (1996), people are more likely to generate and provide unusual, useful ideas if they are either legitimated to do so by the situation or explicitly instructed. Since

ideas usually originate from individuals, but innovation is typically the result of a joint effort of several different individuals, innovation team members must be willing to develop ideas into innovation. Crucial for innovation success then is to trigger action thresholds of individual innovation team members for appreciating and being attentive to new ideas and opportunities (van de Ven, 1986). When management is perceived as not being supportive of innovative behaviour, participants in the innovation process may not be willing to support innovative ideas. The underlying rationale for such a behaviour could refer to team members' assumption that those ideas can hardly survive without sufficient management support (Janssen, 2005). Nevertheless, management needs to employ the right mix of support and criticism. As van de Ven and Grazman (1997) observed, too much support and not enough criticism may result in an innovation project that, for instance, proceeds through the innovation process but finally fails in the market. Conversely, too much criticism may discourage the innovation team to produce enough ideas.

Altogether, management needs to support entrepreneurship, encourages employees to take the initiative, and needs to provide a clear vision to guide innovation efforts (Swink, 2000; van de Ven and Chu, 1989; de Brentani and Kleinschmidt, 2004; Kleinschmidt et al., 2007). However, a supporting and encouraging management is a prerequisite for any innovation culture – regardless of the underlying innovation strategy, i.e., open or closed. Therefore, we do not expect any differences between Open Innovation units and Closed Innovation units regarding the underlying management support for innovative behaviour. Hence, the following hypothesis is proposed:

- Hypothesis 3 Open Innovation units and Closed Innovation units are not culturally different regarding the degree of management support of innovative behaviour.

4 Research design and measures

4.1 Research setting – the chemical industry

Our study focuses on the chemical industry. The chemical industry is one of the largest, most complex and most diversified industries at all. Due to the enormous diversity of product, the chemical industry supplies virtually into every other industry. Having its starting point in the petrochemicals sector, chemicals form the building blocks at every level of production and consumption in many industries (Arora et al., 1998). The last century was undoubtedly the century of chemical innovation. Natural products like paper, wood, or cotton were replaced by synthetic substances. For instance, the big fibre innovations nylon, polyester and acrylic tremendously changed many industrial and consumer markets. They systematically substituted the traditional natural cotton and additionally made totally new applications possible (Landau, 1998). However, in the last 20 years, the flow of major breakthroughs in chemical molecules dried up. Far fewer really new molecules were developed or discovered and changes in properties of existing molecules – in comparison with those of existing substances – became incremental (Schlenzka and Meffert, 2001). At the same time, however, radically new inventions and technology platforms which are of high importance to the chemical industry have been developed in emerging industry sectors, such as biotechnology or nanotechnology.

Especially in these areas, which are crucial for long-term competitiveness, relevant competencies are developed outside the big players. Here, specialised small and medium-sized enterprises as well as academic research play a dominant role (Schlenzka and Meffert, 2001). Therefore, a more open approach to innovation appears to be necessary.

This pressing need to integrate external ideas, technologies, and knowledge sources in innovation processes has been recognised by many players in the chemical industry. Accordingly, they applied an open innovation approach to a certain extent by setting up separated organisational units which are responsible for exploring innovation (e.g., Evonik's 'Creavis Technologies & Innovation', Celanese's 'Ventures', DSM 'Venturing & Business Development', BASF's 'Joint Innovation Lab', and Bayer's 'Innovation'). Generally, these organisational units focus on innovation projects that lie outside the firm's core businesses or cannot be operated by the internal research departments alone. They usually aim at the development of system solutions that have the potential to extend the product portfolio beyond the pure supply of chemicals, could open up new business areas, or could give access to new customers. Simultaneously, the other organisational units of those firms are further on responsible for developing incremental innovations (Bröring and Herzog, 2008).

4.2 Data collection and measures

As noted by Hofstede et al. (1990, p.289), "[a] crucial question is what represents 'an organisation' from a cultural point of view." Since an organisation may consist of several different departments that are culturally different, it is both a theoretical and empirical problem to determine which units are sufficiently homogeneous. Following Hofstede et al. (1990), we chose a pragmatic approach and relied on management's judgement as to whether a unit was culturally homogeneous. Accordingly, the present study focuses on the innovation cultures of three BUs of a leading multinational company – named ChemCo in this study – within the specialty chemicals industry. The rationale for deploying a one-site sampling scheme refers to the fact that a cross-company research approach would not allow controlling a number of broad contextual factors that influence the innovation culture, such as the historical context of the firm. Therefore, studying innovation cultures in open and closed innovation environments and comparing those cultures with each other would hardly yield meaningful results in a cross-company research design. By focusing on different BUs of one firm, it is ensured that all innovation cultures have the same underlying cultural background. Furthermore, a multiple informant approach was chosen. This is mainly due to the fact that culture is a collective characteristic. A single-respondent approach, therefore, would not yield results which can be generalised for the whole organisational unit. Thus, we focus on individuals' perceptions of their innovation work environment within a specific organisational unit.

Decisions on classifying a BU's innovation model being either open or closed were based on the judgements of ChemCo's Chief Technology Officer and the Senior Vice President for Corporate Innovation Management. Accordingly, one BU (BU OI) applies the open innovation concept while the remaining two BUs (BU CI₁ and BU CI₂) follow a rather closed innovation approach. The Open Innovation unit was deliberately set up to follow an open innovation strategy. For example, in contrast to both Closed Innovation units, the Open Innovation unit has the goal of filling potential technological or

market-related competence gaps with external knowledge of technologies whenever possible. To leave the traditional (closed) innovation path, it is also physically separated from both Closed Innovation units. Furthermore, innovation projects of the Open Innovation unit generally have a longer time horizon than those of both Closed Innovation units which conduct rather short- to mid-term projects.

Data collection was undertaken via a standardised questionnaire, which was developed over several stages. First, the literature was reviewed for theoretical models and scale items – depicting the cultural dimensions of interest – that were previously used in other studies. The wording of items was adapted to suit the context of innovation management within ChemCo's BUs. All items – except those taken from Mehrwald (1999) for the NIH syndrome – were translated into German. Next, the questionnaire was pretested with three experts from the field of innovation and technology management and with three employees from ChemCo. The pretest resulted in some minor changes regarding the wording of items (see Table 1). It has to be noted that Mehrwald (1999) conceptualised the NIH syndrome as a multidimensional construct. We only focused on one of those dimensions, i.e., the degree of trust that employees have in the firm's technological competence. This construct originally consists of six items which we – based on the factor loadings reported by Mehrwald – reduced to five items. The revised instrument was administered by mail (open innovation unit OI), personally (Closed Innovation unit CI₁), and by e-mail (Closed Innovation unit CI₂) from July until September 2006. Respondents were approached differently due to personal demands of the respective R&D Vice Presidents. Due to the fact that the names of all potential participants were known and the survey was supported by management, almost all relevant employees within each BU participated. Only in some cases, employees were not available during the data collection period. Overall, extremely high response rates of more than 90% could be realised in each BU. Hence, virtually the entire available sample was measured. A total of 120 employees participated. From those, 51 employees came from BU CI₁, 37 from BU CI₂, and 32 from BU OI. Since corporate culture evolves over time and it also takes some time to recognise its different facets, cases were removed from the sample when the respective employee had been working less than six months in the BU. Accordingly, five cases were removed (two in each BU CI₁ and BU CI₂, as well as one in BU OI). Finally, some questionnaires had missing data, so that those cases were removed. Overall, a sample size of $N = 109$ (46 in BU CI₁, 34 in BU CI₂, 29 in BU OI) remained for the following factor analyses.

5 Results

Construct items were subjected to exploratory factor analysis to assess their unidimensionality (see Table 1). The reliability of the measures was evaluated using Cronbach's alpha (range: 0.77 to 0.90) and item-to-total correlations. Construct unidimensionality was supported for the constructs NIH syndrome and management support. However, for the four-item measure organisational risk-taking which was composed of items borrowed from different studies, factor analysis did not support unidimensionality. Therefore, the item which loaded on a second factor was deleted from further analysis. Construct validity was assessed through average variance extracted (VE). Only the construct NIH syndrome showed a VE just below 0.50. However, we decided not to further purify this measure in order to maintain the richness of the

construct. Discriminant validity of the constructs was assessed by applying the Fornell-Larcker criterion (Fornell and Larcker, 1981). All constructs satisfied this condition. Furthermore, for the constructs NIH syndrome and management support as well as for the overall measurement model, goodness-of-fit measures are acceptable.

Table 1 Culture constructs – items, results of factor analyses, and references

<i>Questionnaire items (scale 1-7; strongly disagree/agree)</i>	<i>Factor loading</i>	<i>Item as identified/applied by</i>
NIH syndrome (Cronbach's $\alpha = 0.81$; VE = 0.47; GFI = 0.985; AGFI = 0.955; RMR = 0.069)		
• We rather develop a technology on our own than buying a pig in a poke.	0.80	Mehrwald (1999)
• We rather develop a technology on our own than being dependent on the technology provider's cooperation in order to understand the external technology.	0.75	Mehrwald (1999)
• Even without using external technology, we can achieve market success.	0.56	Mehrwald (1999)
• External technology sourcing is less attractive to our BU, because we would run the risk of disclosing our technological knowledge during the cooperation with a technology provider.	0.74	Mehrwald (1999)
• Using external technology is an important alternative for technology sourcing within this business unit. (R)	0.54	Mehrwald (1999)
Risk-taking (Cronbach's $\alpha = 0.77$; VE = 0.56)		
• When a person tries something new and fails it, it will be considered disadvantageous for the individual's career. (R)	Item deleted	van de Ven and Chu (1989)
• My BU places high value on taking risks, even if there are occasional mistakes.	0.95	van de Ven and Chu (1989); Calantone et al., (2003)
• Failure is acceptable in this BU, if the effort on the innovation project was good.	0.63	Amabile et al. (1996)
• In this BU, risky activities are common place.	0.61	Calantone et al. (2003)
Management support (Cronbach's $\alpha = 0.90$; VE = 0.69; GFI = 1.000; AGFI = 0.998; RMR = 0.012)		
Management has created an open and innovative culture for our new product development activities by...		
• recognising and rewarding entrepreneurship.	0.79	de Brentani and Kleinschmidt (2004)
• actively encouraging employees to submit new product ideas.	0.81	de Brentani and Kleinschmidt (2004); Amabile et al. (1996)
• placing a high level of trust in individuals.	0.81	van de Ven and Chu (1989)
• encouraging individuals to take the initiative.	0.91	van de Ven and Chu (1989)

Notes: (R) = reverse scored item. VE = average variance extracted. The goodness-of-fit measures for the overall measurement model are GFI = 0.963; AGFI = 0.944; RMR = 0.084.

To test our hypotheses, we only focus on perceptions of individuals that come from R&D or R&D supporting functions. Therefore, we had to remove 12 respondents (e.g., from general management) within closed innovation unit CI₁. Answers of the remaining respondents were subjected to analyses of variance (ANOVA) using the type of BU (open vs. closed) as the independent variable and the dimensions of innovation culture as dependent variables. Table 2 depicts the results of ANOVAs as well as respective means and standard deviations.

Table 2 Results of ANOVA

Dimension of innovation culture	Business unit			ANOVA		
	CI ₁ (N = 34)	CI ₂ (N = 34)	OI (N = 29)	F-test	Post hoc tests	
	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)			
NIH syndrome	4.01 (1.24)	3.70 (1.07)	2.42 (0.94)	17.93 ***	(CI ₁ , CI ₂) > OI	**
Risk-taking	2.96 (1.03)	4.23 (1.21)	5.44 (0.98)	41.56 ***	CI ₁ < CI ₂ < OI	**
Management support	3.87 (1.30)	4.96 (1.16)	4.91 (1.29)	8.12 **	CI ₁ < (CI ₂ , OI)	**

Notes: ** $p < 0.01$, *** $p < 0.001$. S.D. = standard deviation. All constructs satisfy the condition of homogeneity of variance. Post hoc tests were computed at $\alpha = 0.01$ using the Bonferroni procedure which is robust in case of unequal sample sizes.

The results of the ANOVA provide support for Hypothesis 1, which predicts that Open Innovation units are less infected with the NIH syndrome than Closed Innovation units. While BUs CI₁ and CI₂ do not significantly differ regarding their degree of NIH syndrome infection, the Bonferroni post hoc test reveals a significant difference between these units and the Open Innovation unit. Regarding the degree of risk-taking, Hypothesis 2 predicts that Open Innovation units – as perceived by employees – are more aggressive regarding risk-taking than Closed Innovation units. Descriptive statistics and the highly significant result of the ANOVA provide support for Hypothesis 2. Bonferroni post hoc tests indicate that the overall significant F statistic is being driven by significant differences between all pairs of groups. Thus, although it was not hypothesised, there is also a highly significant difference in risk-taking between both Closed Innovation units. Finally, Hypothesis 3 predicts that there is no difference between Open and Closed innovation units in how management supports innovative behaviour. Hypothesis 3, however, is only partially supported. On the one hand, the post hoc test reveals that management support in Closed Innovation unit CI₂ and the Open Innovation unit is not significantly different as Hypothesis 3 predicts. On the other hand, management support in Closed Innovation unit CI₁ is significantly lower compared with both other BUs.

6 Discussion and conclusions

6.1 Overview of findings and implications for theory

Our main objective in this paper was to examine innovation cultures of Open and Closed innovation units. To that end, we explored employees' attitudes and perceptions

regarding the NIH syndrome, organisational risk-taking and management support for innovative behaviour.

As it has been shown above, the Open Innovation unit is less infected with the NIH syndrome than each Closed Innovation unit. However, the question remains whether there is any infection at all. To answer this question, we will compare the present study's results with those reported by Mehrwald (1999). Mehrwald (1999) noted that a maximum score of 7.0 on the NIH scale does not clearly indicate an infection with the NIH syndrome. At best, a person that scores 7.0 on the scale could be said to be suspicious of being infected with the NIH syndrome. In general, high scores refer to negative attitudes to external technology. The threshold level is 4.0. Scores less than 4.0 are, in general, classified as positive attitudes, whereas scores greater than 4.0 are considered negative attitudes. Applying this interpretation scheme, no BU is infected with the NIH syndrome. Employees within the Open Innovation unit have a very positive attitude to external technology. For example, those employees do not solely rely on internal technological competencies. Rather, external technology is considered an important alternative for technology sourcing. Employees also think that external technology is needed in order to achieve market success. In contrast, employees within both Closed Innovation units think that they can be successful without using external technology. Overall, however, R&D staff in these two units has a neutral (Closed Innovation unit CI₁) or slightly positive (Closed Innovation unit CI₂) attitude to external technology. This finding is consistent with the findings reported by Mehrwald (1999), who also observed a neutral and even rather positive attitude to external technology. For this NIH dimension, Mehrwald (1999) found 78% of R&D managers having scores of 4.0 or less on the measurement scale. Taking into account that the scale of the NIH syndrome consists of five instead of six items in the present study, Mehrwald (1999) reports an average value of this construct of 3.34. This is similar to the value observed in Closed Innovation unit CI₂.

Overall, the results for the NIH syndrome are consistent with the findings by Mehrwald (1999). That is, at least some warnings of being infected with the NIH syndrome could be found. However, it has to be stressed that Mehrwald's (1999) results are rather consistent with the findings for both Closed Innovation units, especially for Closed Innovation unit CI₂. But why did neither Mehrwald's (1999) results nor the results of this study provide evidence for severe infections with the NIH syndrome? One reason could be that the NIH syndrome has no meaning in corporate practice. However, for more than 30% of the respondents Mehrwald (1999) found at least some warnings of being infected with the NIH syndrome. Hence, the problem does exist in managerial practice. Another reason for not finding severe infections with the NIH syndrome can possibly be seen in the industry focus adopted in Mehrwald (1999) as well as in this study. Accordingly, it has been found by Laursen and Salter (2006) that chemical companies – compared with companies of other industries – follow a relatively open approach to innovation. Since chemical companies have always collaborated with universities and research institutes to a certain extent, they exhibit a higher degree of openness to external knowledge sources.

Interestingly, Mehrwald's (1999) sample consisted of a large amount of respondents from R&D-intensive industries (e.g., engineering or electronics). Among those, the chemical and pharmaceutical industry represented the second largest subgroup. Unfortunately, he does not report results for these subgroups. Therefore, one can only presume that the rather open approach to innovation in the chemical industry has influenced the average degree of the NIH syndrome in Mehrwald's (1999) study.

However, although the Closed Innovation units in this study are not infected with the NIH syndrome, the open Innovation unit still has a significantly more positive attitude towards external technology sourcing. Thus, taking into account industry specifics, one can say that Open Innovation units differ from Closed Innovation units regarding the degree of the NIH syndrome. If this attitude is the ideal one cannot be answered. As Lichtenthaler and Ernst (2006) noted, there may also be a too positive attitude leading to a pure focus on external technology. Since the average value of 2.42 is not at the extreme anchor of the scale, we do not regard the Open Innovation unit being too positive towards external technology.

The second cultural dimension that we considered is organisational risk-taking. As expected, the Open Innovation unit is more open towards risk-taking than both Closed Innovation units. It is further much more willing to take risks. This can be attributed to the fact that the Open Innovation unit ventures into new markets and technologies which is usually associated with higher risks. This finding is also consistent with the open approach to innovation. However, although the Open Innovation unit was expected to exhibit a significantly higher level of risk tolerance compared with the Closed Innovation units, a significant difference between both Closed Innovation units was not expected. Employees in Closed Innovation unit CI₁ perceive their work environment as being risk averse. Participants particularly scored low on those items that ask about the degree of risk-taking being valued and the degree of risky activities being common place. One possible explanation addresses the underlying R&D time horizon. Since Closed Innovation unit CI₁ develops products and technologies that are supposed to be commercialised within the near future, taking too many risks could jeopardise its market position. The same, however, applies to Closed Innovation unit CI₂ which is rather risk neutral. Thus, it could be assumed that risk-taking in Closed Innovation unit CI₂ is not biased at all in that rational and well-considered decisions with regard to market and technology risks are made. Furthermore, it needs to be questioned if the degree of organisational risk-taking in Closed Innovation unit CI₁ is conducive to innovative performance. One may even assume that this practice of risk-taking does not correspond to an innovation culture.

The third cultural dimension in this study is management support of innovative behaviour. Although management support of innovative behaviour is significantly weaker in Closed Innovation unit CI₁ compared to the other BUs, it can still be regarded as being rather neutral. That is, management neither excessively fosters nor inhibits entrepreneurship. In contrast, the other two BUs are characterised by a management that also avoids excessive encouragement of innovative behaviour, but rather follows a modest approach of encouraging and supporting people. Although we did not examine the influence of management support on employees' willingness to submit new product or technology ideas, etc., we assume that at least in Closed Innovation unit CI₂ and the Open Innovation unit employees are not distracted from innovative behaviour.

Regarding research on open innovation, our study highlights the differences in innovation cultures between BUs following an open innovation strategy and those following a closed innovation strategy. This is not to say, for instance, that innovation team members should have a strong positive attitude towards external technology as we found in the Open Innovation unit. It should always be contingent on other factors related to innovation as well, such as the appropriability regime or the degree of technological or market turbulence, if being infected with the NIH syndrome is valued negatively or positively. With respect to the innovation strategy followed, we consider all three BUs

showing adequate levels of the NIH syndrome. In a similar manner, we cannot state that taking greater risks will result in higher innovative performance, although some researchers have argued that risk-taking increases, for instance, NPD speed (Calantone et al., 2003). Again, other factors have to be considered. Calantone et al. (2003) found that risk-taking is particularly advantageous in case of high technological turbulence. However, we think that with respect to the underlying innovation strategy, all three BUs manage risk effectively except Closed Innovation unit CI₁. Here, management should be cautious of being too risk averse which could cause paralysis in decision-making. Regarding the management support of innovative behaviour, we assume contingency factors playing a less important role. Innovation always requires people that take the initiative, submit unusual ideas, or act in an entrepreneurial way. Thus, following either an open or a closed innovation strategy does not require a strategy-specific type of management support. With regard to RBV, it seems that ChemCo had the necessary dynamic capabilities to establish an adequate innovation culture within its Open Innovation unit.

6.2 Managerial implications

In the course of the study, some managerial implications became apparent. The following implications provide guidelines for firms in their attempts to establish an adequate open innovation culture. First of all, it is worthwhile to quantitatively assess innovation culture in order to identify gaps between the existing and the desired state of innovation culture. Regarding the research focus of this study, firms which have set up an Open Innovation unit can monitor the success of such an endeavour. This becomes particularly relevant when firms implement the open innovation concept in order to create new businesses or develop radical innovations. Since both are typically long-term endeavours, quantitative culture assessment of such a 'new' organisational unit offers an attractive tool to measure the impact and/ or progress of open innovation initiatives. For example, since external technology sourcing is important to fully embark on the open innovation model, the NIH syndrome is detrimental to the open innovation success. Assessing the degree of infection with the NIH syndrome could yield important insights (considerably) in advance of any negative outcomes. Thus, innovation culture assessment is a very useful indicator for future innovation success. Managers can therefore early create awareness of those differences between the actual and the desired state of open innovation culture.

When the existing state of an innovation culture deviates from the desired state and management wants to change innovation culture, culture change efforts should take into account the different cultural levels of shared basic values, norms, and practices. Each level requires a different approach of culture change management. On the cultural level of norms, the NIH syndrome has been studied. To reduce or avoid the NIH syndrome, management should create awareness among employees by confronting them with the results of the quantitative assessment. Furthermore, management should persuasively communicate other persons' or firms' positive experiences with external technology sourcing to its R&D staff. Different incentive schemes that reward R&D staff based on its ability to solve technical problems – regardless of the source (internal or external) of the solution – also provide an adequate means to influence behaviour. Admittedly, those incentives and directives not necessarily lead to a change in employees' attitudes in the short run. However, employees may change their attitudes after positive experiences with using external technologies and external pathways to market. Such a change in attitudes

is likely to have a long-term and sustainable effect on the success of an open innovation endeavour. It needs to be stressed, however, that determining the ideal degree of the NIH syndrome is difficult to define since it is highly interdependent with the underlying innovation environment and the existing competencies of the firm. Firms could therefore define specific intervals for employees' attitudes towards external technology sourcing.

Closely related to the discussion of changing incentives and employing adequate directives is the fact that employees need to perceive their management as being open to external technology sourcing. This refers to the surface level of innovation culture, the level of practices. Managers should work directly on this level, since it is, on the one hand, much more directly amenable and, on the other hand, much more observable and likely to be perceived by employees. From this perspective, the present study suggests that managers should place high levels of trust in individuals and encourage employees to take the initiative even with the risk of failure.

6.3 Limitations and future research opportunities

Finally, some limitations should be emphasised, which also provide further research opportunities. First, the ability to generalise the findings of this study to other firms and industries was traded for an increase of control over firm and industry specific variations by collecting data only for different BUs of a single firm. A potential limitation is that data was only collected for different BUs of one firm. On the one hand, this one-site sampling approach ensures that the analysed BUs are all equally affected by the same historical context of the overall firm. On the other hand, the present study's results may or may not generalise to other firms in other industries or even in the same industry. Future research could therefore investigate and compare innovation cultures of other firms in the chemical industry. As several firms in the chemical industry established separated organisational units that follow an open innovation strategy, these units could be contrasted to units innovating in a closed innovation mode. Furthermore, it would be of interest if other industries show similar differences in innovation culture between open and closed organisational units. Another limitation is that we excluded contingencies other than innovation strategy. For instance, environmental turbulence (market and/or technology) might be used to assess the stability of the results. Also, employees' personalities could be included as a covariate in order to account for differences of, for example, individuals' self-confidence or risk aversion which is likely to influence their perceptions. Finally, open innovation is a relatively new concept in technology and innovation management, which is not yet fully implemented by the majority of firms. Since the use of open innovation is expected to further increase in the future, the present study's findings, which are based on the current situation of open innovation in practice, may have to be reconsidered in the future if more firms have implemented the concept and are more experienced with using it. The successful use of external technologies within Open Innovation units may, for example, trigger opening up corporate boundaries in former Closed Innovation units. In general, however, the results may be expected to hold true in the long-term.

Despite these limitations and future research opportunities, we believe that the work presented makes a worthwhile contribution by shedding new light on the people side of the equation in open and closed innovation environments.

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